

TP github Devops /lprec Cédric Pourret - Avril 2025

Effectué sur VM Ubuntu server 22.04 créée avec proxmox Commandes passées par SSH depuis Moba Xterm

```
13/04/2025 15:08.44 /home/mobaxterm ssh access@192.168.1.172
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-136-generic x86_64)
```

```
* Documentation: https://help.ubuntu.com
* Management:   https://landscape.canonical.com
* Support:       https://ubuntu.com/pro
```

System information as of dim. 13 avril 2025 13:08:53 UTC

```
System load:          0.0
Usage of /:            50.8% of 9.75GB
Memory usage:         12%
Swap usage:           0%
Processes:            111
Users logged in:      0
IPv4 address for ens18: 192.168.1.172
IPv6 address for ens18: 2a02:842b:8462:6a01:be24:11ff:fea6:6fa4
```

La maintenance de sécurité étendue pour Applications n'est pas activée.

54 mises à jour peuvent être appliquées immédiatement.

Pour afficher ces mises à jour supplémentaires, exécuter : `apt list --upgradable`

Enable ESM Apps to receive additional future security updates.

See <https://ubuntu.com/esm> or run: `sudo pro status`

[→](#) [↺](#) [🔒](#) [🔗](#) <https://github.com/new> [🔍](#) [☆](#) [📧](#) [⬇️](#) [👤](#) [📁](#) [📄](#)

Create a new repository

A repository contains all project files, including the revision history. Already have a project repository elsewhere? [Import a repository.](#)

Required fields are marked with an asterisk ().*

Repository template

No template ▾

Start your repository with a template repository's contents.

Owner *

Repository name *

 cedret ▾

/

github2cp

✅ github2cp is available.

Great repository names are short and memorable. Need inspiration? How about **cuddly-bassoon** ?

Description (optional)

☐

 **Public**
Anyone on the internet can see this repository. You choose who can commit.

☒

 **Private**
You choose who can see and commit to this repository.

Initialize this repository with:

☐ Add a README file
This is where you can write a long description for your project. [Learn more about READMEs.](#)

Add .gitignore

.gitignore template: None ▾

Choose which files not to track from a list of templates. [Learn more about ignoring files.](#)

Choose a license

License: None ▾

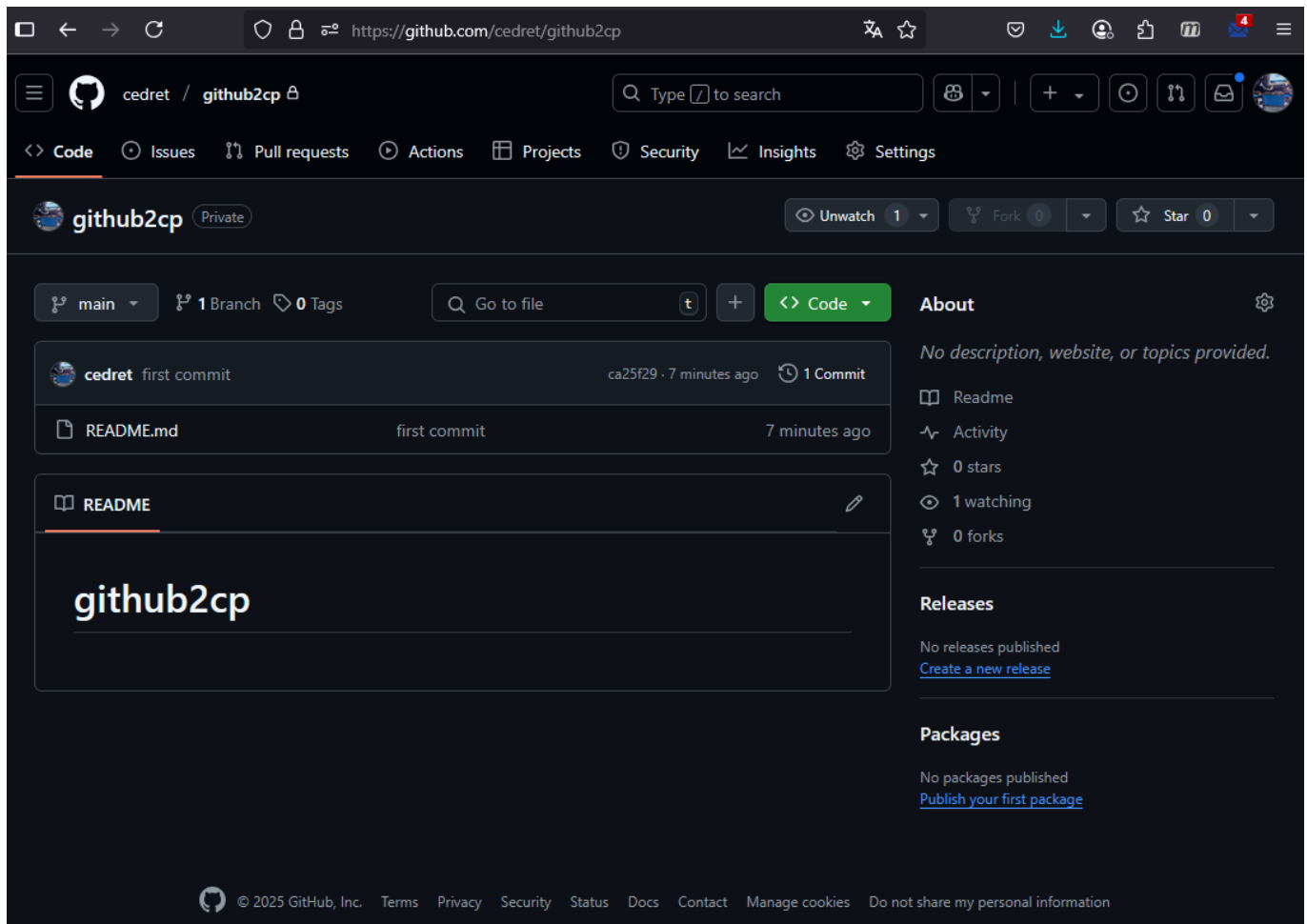
A license tells others what they can and can't do with your code. [Learn more about licenses.](#)

 You are creating a private repository in your personal account.

Create repository

Création du repository selon Github

```
access@devops1iprec:~$ mkdir github2
access@devops1iprec:~$ cd github2
access@devops1iprec:~/github2$ echo "# github2cp" >> README.md
access@devops1iprec:~/github2$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/access/github2/.git/
access@devops1iprec:~/github2$ git add README.md
access@devops1iprec:~/github2$ git commit -m "first commit"
[master (root-commit) ca25f29] first commit
 1 file changed, 1 insertion(+)
 create mode 100644 README.md
access@devops1iprec:~/github2$ git branch -M main
access@devops1iprec:~/github2$ git remote add origin
git@github.com:cedret/github2cp.git
access@devops1iprec:~/github2$ git push -u origin main
Enter passphrase for key '/home/access/.ssh/id_ed25519':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 212 bytes | 212.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:cedret/github2cp.git
 * [new branch]      main -> main
Branch 'main' set up to track remote branch 'main' from 'origin'.
access@devops1iprec:~/github2$ git remote -v
origin  git@github.com:cedret/github2cp.git (fetch)
origin  git@github.com:cedret/github2cp.git (push)
access@devops1iprec:~/github2$ ls -al
total 16
drwxrwxr-x  3 access access 4096 avril 13 16:21 .
drwxr-x--- 13 access access 4096 avril 13 16:20 ..
drwxrwxr-x  8 access access 4096 avril 13 16:22 .git
-rw-rw-r--  1 access access  12 avril 13 16:21 README.md
```



Script tableau de langues

```
access@devops1iprec:~/github2$ mkdir scripts
access@devops1iprec:~/github2$ cd scripts/
access@devops1iprec:~/github2/scripts$ sudo nano langues.sh
[sudo] password for access:
access@devops1iprec:~/github2/scripts$ sudo chmod +x langues.sh
access@devops1iprec:~/github2/scripts$ cat langues.sh
```

```
#!/bin/bash

blanc="\033[0m"
jaune="\033[1;33m"

# Déclaration d'un tableau de langues
langues=("Français" "Anglais" "Espagnol" "Allemand" "Italien" "Japonais")

while true;
do
    echo -e "${jaune}Choisir:"
    echo "11 pour afficher tableau"
    echo "12 pour choisir un élément"
    echo "13 pour modifier un élément"
```

```
echo "14 pour ajouter un élément"
echo "15 pour effacer le tableau"
echo "0 pour quitter"
read reponse
case $reponse in
    0)
        echo -e "${blanc}fin"
        break
        ;;
    11)
        # Affichage du tableau
        echo "Liste des langues :"
        for langue in "${langues[@]}"
        do
            echo "- $langue"
        done
        ;;
    12)
        echo "quel numéro d'élément afficher svp?"
        read affiche
        # Affichage l'élément (indice affiche-1)
        echo "L'élément $affiche du tableau est : ${langues[$affiche-1]}"
        ;;
    13)
        echo "quel numéro d'élément remplacer svp?"
        read remplace
        echo "Quelle langue ajouter svp?"
        read langue
        echo "$langue va être remplacé(e)"
        langues[$remplace-1]=$langue
        # Affichage l'élément (indice remplace-1)
        echo "L'élément $remplace du tableau est maintenant:
${langues[$remplace-1]}"
        ;;
    14)
        echo "Quelle langue ajouter svp?"
        read ajoute
        echo "$ajoute est ajoutée au tableau"
        langues+=("$ajoute")
        ;;
    15)
        echo "99 pour effacer le tableau"
        read efface
        if [ $efface -eq 99 ]
        then
            langues=()
        fi
        ;;
esac

done
```

Script tranches d'âge

```
access@devops1iprec:~/github2/scripts$ sudo nano age.sh
access@devops1iprec:~/github2/scripts$ chmod +x age.sh
chmod: changing permissions of 'age.sh': Operation not permitted
access@devops1iprec:~/github2/scripts$ sudo chmod +x age.sh
access@devops1iprec:~/github2/scripts$ cat age.sh
#!/bin/bash

while true;
do
    read -p "Entrez l'âge de la personne: " age

    # Vérification que l'âge est un nombre positif
    if ! [[ "$age" =~ ^[0-9]+$ ]]; then
        echo "Veuillez entrer un nombre entier positif."
        echo "Fin du programme"
        exit 1
    break
    fi

    # Déterminer la tranche d'âge
    if [ "$age" -lt 18 ]; then
        echo "Tranche d'âge : Enfant"
    elif [ "$age" -lt 64 ]; then
        echo "Tranche d'âge : Adulte"
    elif [ "$age" -lt 130 ]; then
        echo "Tranche d'âge : Senior"
    else
        echo "Inhumain"
        break
    fi
done
```

Script nombre inférieur à 20

```
access@devops1iprec:~/github2/scripts$ sudo nano nombre.sh
access@devops1iprec:~/github2/scripts$ chmod +x nombre.sh
chmod: changing permissions of 'nombre.sh': Operation not permitted
access@devops1iprec:~/github2/scripts$ sudo chmod +x nombre.sh
access@devops1iprec:~/github2/scripts$ cat nombre.sh

#!/bin/bash

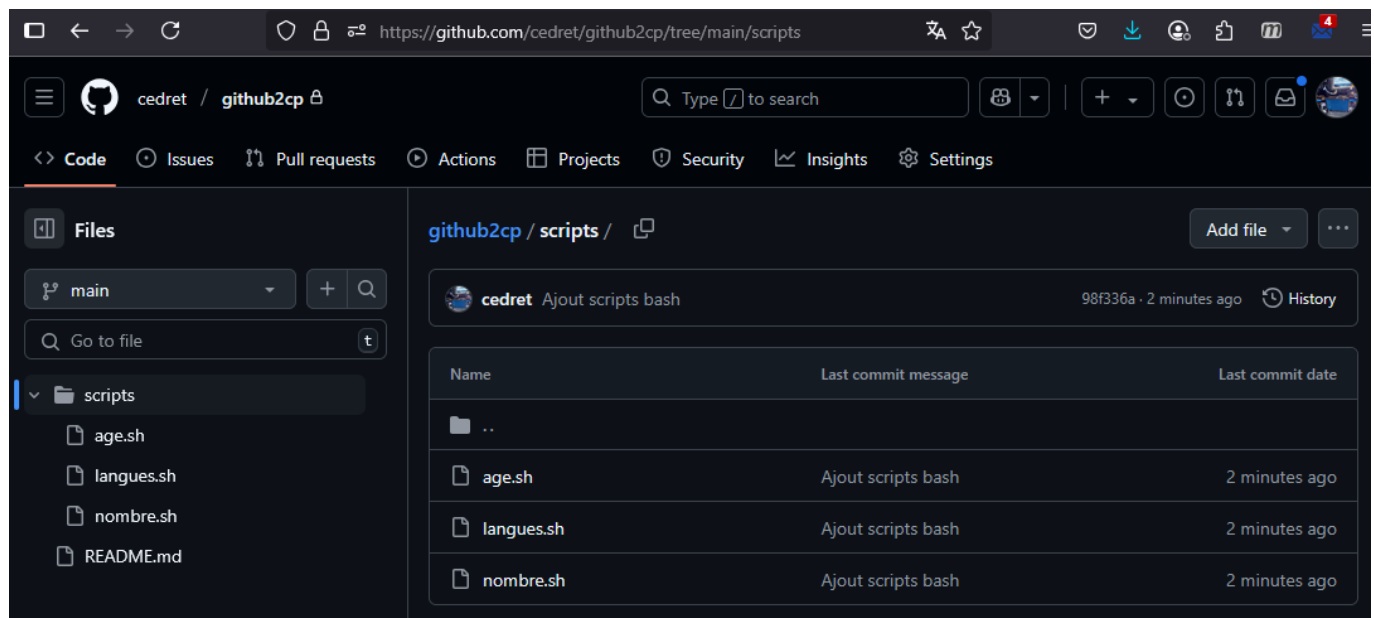
# Boucle tant que le nombre est inférieur ou égal à 20
nombre=0
cycle=0
while [ "$nombre" -le 20 ]; do
    read -p "Entrez un nombre supérieur à 20 : " nombre
    ((cycle+=1))
```

```
done
```

```
echo "Merci ! Vous avez entré : $nombre en $cycle tentative(s)"
```

Mise à jour Github

```
access@devops1iprec:~/github2/scripts$ cd ..
access@devops1iprec:~/github2$ git add .
access@devops1iprec:~/github2$ git commit -m "Ajout scripts bash"
[main 98f336a] Ajout scripts bash
 3 files changed, 103 insertions(+)
 create mode 100755 scripts/age.sh
 create mode 100755 scripts/langues.sh
 create mode 100755 scripts/nombre.sh
access@devops1iprec:~/github2$
access@devops1iprec:~/github2$ git push origin main
Enter passphrase for key '/home/access/.ssh/id_ed25519':
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 1.45 KiB | 1.45 MiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:cedret/github2cp.git
   ca25f29..98f336a  main -> main
```



The screenshot shows the GitHub web interface for the repository 'cedret / github2cp'. The 'Code' tab is selected, and the 'scripts' directory is viewed. The commit history for the 'main' branch is displayed, showing a single commit with the message 'Ajout scripts bash' and hash '98f336a' from 2 minutes ago. The table below lists the files added in this commit.

Name	Last commit message	Last commit date
..		
age.sh	Ajout scripts bash	2 minutes ago
langues.sh	Ajout scripts bash	2 minutes ago
nombre.sh	Ajout scripts bash	2 minutes ago

Changement de machine pour récupération de repository et création du workflow Github actions

Ouverture session

```

□ 13/04/2025 □ □ 18:42.28 □ □ /home/mobaxterm □ ssh access@192.168.1.144
Warning: Permanently added '192.168.1.144' (ED25519) to the list of known hosts.
access@192.168.1.144's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-136-generic x86_64)

```

```

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

```

System information as of dim. 13 avril 2025 16:42:38 UTC

```

System load:            0.07
Usage of /:              48.8% of 9.75GB
Memory usage:           11%
Swap usage:              0%
Processes:               117
Users logged in:         0
IPv4 address for ens18: 192.168.1.144
IPv6 address for ens18: 2a02:842b:8462:6a01:be24:11ff:fe27:c8b3

```

La maintenance de sécurité étendue pour Applications n'est pas activée.

54 mises à jour peuvent être appliquées immédiatement.
 Pour afficher ces mises à jour supplémentaires, exécuter : `apt list --upgradable`

Enable ESM Apps to receive additional future security updates.
 See <https://ubuntu.com/esm> or run: `sudo pro status`

The programs included with the Ubuntu system are free software;
 the exact distribution terms for each program are described in the
 individual files in `/usr/share/doc/*/copyright`.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
 applicable law.

```

/usr/bin/xauth:  file /home/access/.Xauthority does not exist
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

```

Création clé SSH pour connexion

```

access@iprecgithub:~/github2local$ ssh-keygen -t ed25519 -C "cedret7@gmail.fr"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/access/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/access/.ssh/id_ed25519
Your public key has been saved in /home/access/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:80Tmy6Q0hs0J2oCsp0SvbmNzg6QSTjysfwA/7Xn02nU cedret7@gmail.fr

```


The key's randomart image is:

```
+-[ED25519 256]--+
```

```
|
|
|      . .
|o.      =
|+*...  S
|B*++.=.+
|Oo++ooB.o . E
|+B.o+o.*.o .
|=+*o..o.+
|
```

```
+----[SHA256]-----+
```

```
access@iprecgithub:~/github2local$ cat
```

```
access@iprecgithub:~/github2local$ cat /home/access/.ssh/id_ed25519.pub
```

```
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIGMFkXhLdg71FnnwO+h12w6m7NMA6A00ruvp1NEkEfwg
cedret7@gmail.fr
```

```
access@iprecgithub:~/github2local$
```

```
access@iprecgithub:~/github2local$ git clone git@github.com:cedret/github2cp.git
Cloning into 'github2cp'...
```

```
Enter passphrase for key '/home/access/.ssh/id_ed25519':
```

```
remote: Enumerating objects: 9, done.
```

```
remote: Counting objects: 100% (9/9), done.
```

```
remote: Compressing objects: 100% (7/7), done.
```

```
remote: Total 9 (delta 0), reused 9 (delta 0), pack-reused 0 (from 0)
```

```
Receiving objects: 100% (9/9), done.
```

The screenshot shows the GitHub Settings page for a user named 'cedret'. The page is divided into a left sidebar with navigation links and a main content area. The sidebar includes links for Public profile, Account, Appearance, Accessibility, Notifications, Access, Billing and plans, Emails, Password and authentication, Sessions, SSH and GPG keys (highlighted), Organizations, Enterprises, Moderation, Code, planning, and automation, Repositories, Codespaces, Packages, Copilot, Pages, Saved replies, and Security. The main content area has a success message at the top: 'You have successfully added the key 'github2devops''. Below this, the 'SSH keys' section shows two keys: 'testartromm' (added Apr 12, 2025, last used within the last week) and 'github2devops' (added Apr 13, 2025, never used). Both keys have a 'Delete' button. The 'GPG keys' section states there are no GPG keys associated with the account. The 'Vigilant mode' section has a checkbox for 'Flag unsigned commits as unverified' which is currently unchecked.

Préparation pour workflows

```
access@iprecgithub:~/github2local$ ls -al
total 12
drwxrwxr-x 3 access access 4096 avril 13 16:52 .
drwxr-x--- 5 access access 4096 avril 13 16:45 ..
drwxrwxr-x 4 access access 4096 avril 13 16:52 github2cp
access@iprecgithub:~/github2local$ cd github2cp/
access@iprecgithub:~/github2local/github2cp$ ls -al
total 20
drwxrwxr-x 4 access access 4096 avril 13 16:52 .
drwxrwxr-x 3 access access 4096 avril 13 16:52 ..
drwxrwxr-x 8 access access 4096 avril 13 16:52 .git
```

```
-rw-rw-r-- 1 access access 12 avril 13 16:52 README.md
drwxrwxr-x 2 access access 4096 avril 13 16:52 scripts

access@iprecgithub:~/github2local/github2cp$ sudo mkdir .github
access@iprecgithub:~/github2local/github2cp$ cd .github/
access@iprecgithub:~/github2local/github2cp/.github$ sudo mkdir workflows
access@iprecgithub:~/github2local/github2cp/.github/workflows$ sudo nano
runbash.yml
access@iprecgithub:~/github2local/github2cp$ cat .github/workflows/runbash.yml
```

Workflow yaml pour scripts

```
name: Exécuter les scripts Bash

on:
  push:
    branches: [ main ]
  workflow_dispatch: # Permet de lancer manuellement le workflow
  pull_request:
    branches: [ main ]

jobs:
  run-bash-scripts:
    runs-on: ubuntu-latest

    steps:
      - name: Récupérer le code
        uses: actions/checkout@v3

      - name: Donner les droits d'exécution aux scripts
        run: chmod +x scripts/*.sh

      - name: Exécuter les scripts Bash
        run: |
          echo " Script: age.sh"
          echo "180" | ./scripts/age.sh

          echo " Script: langues.sh"
          echo "0" | ./scripts/langues.sh

          echo " Script: noombre.sh"
          echo "21" | ./scripts/nombre.sh
```

Mise à jour dépôt

```
access@iprecgithub:~/github2local/github2cp$ ls -al
total 24
drwxrwxr-x 5 access access 4096 avril 13 17:29 .
drwxrwxr-x 3 access access 4096 avril 13 16:52 ..
```

```
drwxrwxr-x 9 access access 4096 avril 13 17:12 .git
drwxr-xr-x 3 root root 4096 avril 13 17:31 .github
-rw-rw-r-- 1 access access 12 avril 13 16:52 README.md
drwxrwxr-x 2 access access 4096 avril 13 16:52 scripts

access@iprecgithub:~/github2local/github2cp$ ls -al .github
total 12
drwxr-xr-x 3 root root 4096 avril 13 17:31 .
drwxrwxr-x 5 access access 4096 avril 13 17:29 ..
drwxr-xr-x 2 root root 4096 avril 13 17:31 workflows

access@iprecgithub:~/github2local/github2cp$ ls -al .github/workflows/
total 12
drwxr-xr-x 2 root root 4096 avril 13 17:31 .
drwxr-xr-x 3 root root 4096 avril 13 17:31 ..
-rw-r--r-- 1 root root 691 avril 13 17:30 runbash.yml
```

```
access@iprecgithub:~/github2local/github2cp$ git add .
access@iprecgithub:~/github2local/github2cp$ git commit -m "ajout wf yaml"
Author identity unknown
*** Please tell me who you are.

Run
  git config --global user.email "you@example.com"
  git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.
fatal: unable to auto-detect email address (got 'access@iprecgithub.(none)')

access@iprecgithub:~/github2local/github2cp$ git config user.name "cedret"
access@iprecgithub:~/github2local/github2cp$ git config user.email
"cedret7@gmail.fr"

access@iprecgithub:~/github2local/github2cp$ git commit -m "ajout wf4yaml"
[main 67d1a55] ajout wf4yaml
 1 file changed, 30 insertions(+)
 create mode 100644 .github/workflows/runbash.yml
access@iprecgithub:~/github2local/github2cp$ git push origin main
Enter passphrase for key '/home/access/.ssh/id_ed25519':
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 712 bytes | 712.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:cedret/github2cp.git
   98f336a..67d1a55  main -> main
access@iprecgithub:~/github2local/github2cp$ git log
commit 67d1a559292314333043cd17d287325e13021c63 (HEAD -> main, origin/main,
origin/HEAD)
```

Author: cedret <cedret7@gmail.fr>
Date: Sun Apr 13 17:36:12 2025 +0000

ajout wf4yaml

commit 98f336a6e0c85728779759af497d5220bb064d6b
Author: cedret <cedret7@gmail.fr>
Date: Sun Apr 13 16:31:28 2025 +0000

Ajout scripts bash

commit ca25f29ac28e90d1f8f6cd8bd203f3da841d769e
Author: cedret <cedret7@gmail.fr>
Date: Sun Apr 13 16:22:03 2025 +0000

first commit

The screenshot shows a GitHub repository page for 'cedret / github2cp'. The left sidebar displays the file structure with folders '.github/workflows' and 'scripts', and files 'age.sh', 'langues.sh', 'nombre.sh', and 'README.md'. The main content area shows the workflow file 'runbash.yml' with a commit history of 1 run. The workflow is titled 'Exécuter les scripts Bash' and is triggered on push to the 'main' branch. It includes a job 'run-bash-scripts' that runs on 'ubuntu-latest' and performs the following steps: checkout code, set permissions for scripts, and execute three scripts: 'age.sh', 'langues.sh', and 'nombre.sh'.

```
1  name: Exécuter les scripts Bash
2
3  on:
4    push:
5      branches: [ main ]
6    workflow_dispatch: # Permet de lancer manuellement le workflow
7    pull_request:
8      branches: [ main ]
9
10 jobs:
11   run-bash-scripts:
12     runs-on: ubuntu-latest
13
14   steps:
15     - name: Récupérer le code
16       uses: actions/checkout@v3
17
18     - name: Donner les droits d'exécution aux scripts
19       run: chmod +x scripts/*.sh
20
21     - name: Exécuter les scripts Bash
22       run: |
23         echo " Script: age.sh"
24         echo "180" | ./scripts/age.sh
25
26         echo " Script: langues.sh"
27         echo "0" | ./scripts/langues.sh
28
29         echo " Script: noombre.sh"
30         echo "21" | ./scripts/nombre.sh
```

The screenshot shows the GitHub Actions interface for a workflow named 'ajout wf4yaml #1'. The workflow is triggered by a push to the 'main' branch. The status is 'Success' with a total duration of 8s. The workflow file is 'runbash.yml'. The jobs section shows a single job 'run-bash-scripts' which is also successful and took 4s to complete. The left sidebar contains links to 'Summary', 'Jobs', 'Run details', 'Usage', and 'Workflow file'.

Modification YAML

```
access@iprecgithub:~/github2local/github2cp$ sudo nano
.github/workflows/runbash.yml
access@iprecgithub:~/github2local/github2cp$ cat .github/workflows/runbash.yml

name: Exécuter les scripts Bash

on:
  push:
    branches: [ main ]
  workflow_dispatch: # Permet de lancer manuellement le workflow
  pull_request:
    branches: [ main ]

jobs:
  run-bash-scripts:
    runs-on: ubuntu-latest

    steps:
      - name: Récupérer le code
        uses: actions/checkout@v3

      - name: Donner les droits d'exécution aux scripts
        run: chmod +x scripts/*.sh

      - name: Exécuter les script age
```

```

run: |
    echo " Script: age.sh"
    echo "180" | ./scripts/age.sh

- name: Exécuter le script langue
  run: |
    echo " Script: langues.sh"
    echo "0" | ./scripts/langues.sh

- name: Exécuter le script nombre
  run: |
    echo " Script: nombre.sh"
    echo "11" > test.txt
    echo "21" >> test.txt
    ./scripts/nombre.sh < test.txt

```

Mise à jour sur Github

```

access@iprecgithub:~/github2local/github2cp$ git add .
access@iprecgithub:~/github2local/github2cp$ git commit -m "ajout wf5yaml"
[main dfc9e9a] ajout wf5yaml
 1 file changed, 9 insertions(+), 3 deletions(-)
access@iprecgithub:~/github2local/github2cp$ git push origin main
Enter passphrase for key '/home/access/.ssh/id_ed25519':
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 518 bytes | 518.00 KiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:cedret/github2cp.git
 67d1a55..dfc9e9a  main -> main
access@iprecgithub:~/github2local/github2cp$ git log
commit dfc9e9a4814a1d49c7acfa68425d5a879d9c2b68 (HEAD -> main, origin/main,
origin/HEAD)
Author: cedret <cedret7@gmail.fr>
Date:   Sun Apr 13 17:48:33 2025 +0000

    ajout wf5yaml

commit 67d1a559292314333043cd17d287325e13021c63
Author: cedret <cedret7@gmail.fr>
Date:   Sun Apr 13 17:36:12 2025 +0000

    ajout wf4yaml

commit 98f336a6e0c85728779759af497d5220bb064d6b
Author: cedret <cedret7@gmail.fr>
Date:   Sun Apr 13 16:31:28 2025 +0000

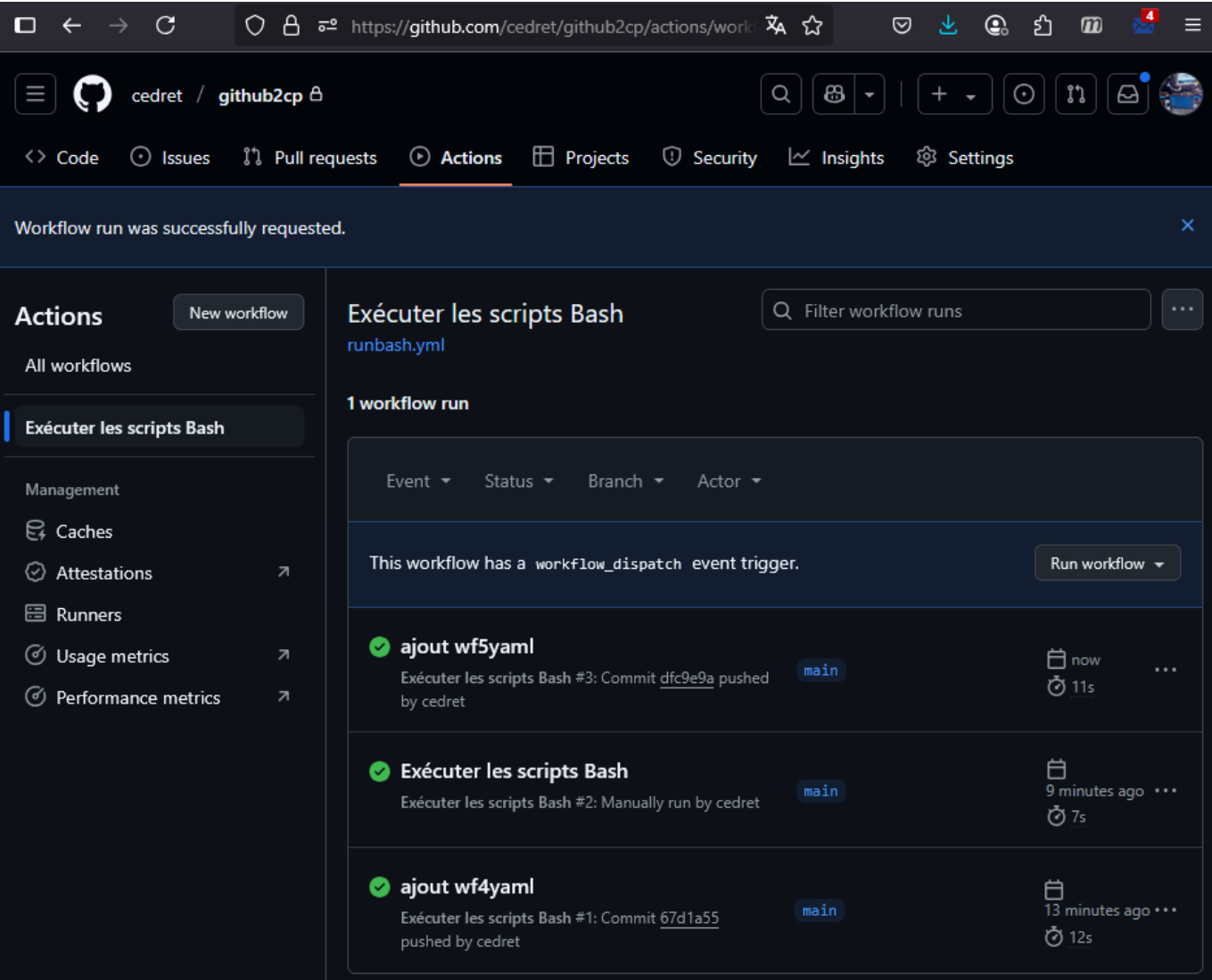
```

```
Ajout scripts bash

commit ca25f29ac28e90d1f8f6cd8bd203f3da841d769e
Author: cedret <cedret7@gmail.fr>
Date:   Sun Apr 13 16:22:03 2025 +0000

first commit
```

Il s'est déclenché automatiquement



Fin